Sahdev Singh | MCA (GenAI)

Assignment (Module 1): Core Java

Task 1:

Write a program that declares variables for a student's name, roll number, and marks in three subjects. Assign values and print them. Ensure all variable names follow Java identifier rules.

Solution -

```
public class StudentDetails {
    public static void main(String[] args) {

        String studentName = "Rahul Sharma";
        int rollNumber = 101;
        int marksSubject1 = 85;
        int marksSubject2 = 90;
        int marksSubject3 = 88;

        // Printing the details
        System.out.println("Student Name: " + studentName);
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Marks in English: " + marksSubject1);
        System.out.println("Marks in Math: " + marksSubject2);
        System.out.println("Marks in Physics: " + marksSubject3);
    }
}
```

StudentDetails.java

Java > StudentDetails.java

```java
public class StudentDetails {
    Run | Debug
    public static void main(String[] args) {

        String studentName = "Rahul Sharma";
        int rollNumber = 101;
        int marksSubject1 = 85;
        int marksSubject2 = 90;
        int marksSubject3 = 88;

        // Printing the details
        System.out.println("Student Name: " + studentName);
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Marks in English: " + marksSubject1);
        System.out.println("Marks in Math: " + marksSubject2);
        System.out.println("Marks in Physics: " + marksSubject3);
    }
}
```

Problems 4 | Output | Debug Console | Terminal | Ports

```
PS D:\Programs and Projects>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExc
eptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a65272379
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'StudentDetails'
Student Name: Rahul Sharma
Roll Number: 101
Marks in English: 85
Marks in Math: 90
Marks in Physics: 88
PS D:\Programs and Projects>
```

Task 2:

Write a Java program that:
• Stores an integer value for temperature in Celsius.
• Converts it to Fahrenheit using the formula F = (C × 9/5) + 32.
• Displays both Celsius and Fahrenheit values.

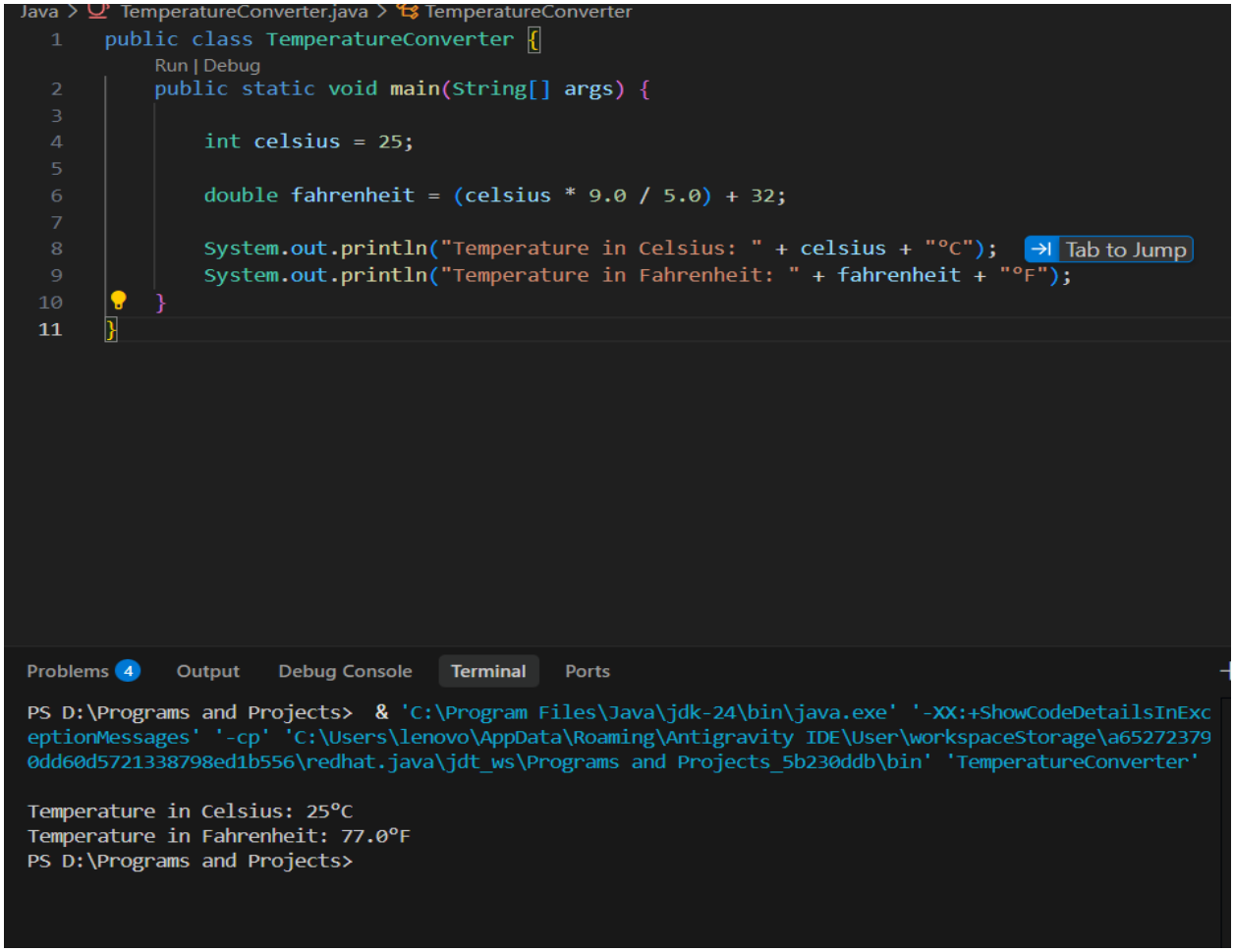
Solution -

```
public class TemperatureConverter {
    public static void main(String[] args) {

        int celsius = 25;

        double fahrenheit = (celsius * 9.0 / 5.0) + 32;

        System.out.println("Temperature in Celsius: " + celsius + "°C");
        System.out.println("Temperature in Fahrenheit: " + fahrenheit +
"°F");
    }
}
```


Java > TemperatureConverter.java > TemperatureConverter
public class TemperatureConverter {
Run | Debug
public static void main(String[] args) {
int celsius = 25;
double fahrenheit = (celsius * 9.0 / 5.0) + 32;
System.out.println("Temperature in Celsius: " + celsius + "°C");
Tab to Jump
System.out.println("Temperature in Fahrenheit: " + fahrenheit + "°F");
}
}
Problems 4
Output
Debug Console
Terminal
Ports
PS D:\Programs and Projects> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a652723790dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'TemperatureConverter'
Temperature in Celsius: 25°C
Temperature in Fahrenheit: 77.0°F
PS D:\Programs and Projects>

Task 3:
Accept two integers from the user and perform addition, subtraction, multiplication, division, and modulus. Display results for each operation.

Solution -

```
import java.util.Scanner;

public class Calculator {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.println("Addition = " + (a + b));
        System.out.println("Subtraction = " + (a - b));
        System.out.println("Multiplication = " + (a * b));
        System.out.println("Division = " + (a / b));
        System.out.println("Modulus = " + (a % b));
    }
}
```

Calculator.java 1

Java > Calculator.java > Calculator

```java
import java.util.Scanner;

public class Calculator {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.println("Addition = " + (a + b));
        System.out.println("Subtraction = " + (a - b));
        System.out.println("Multiplication = " + (a * b));
        System.out.println("Division = " + (a / b));
        System.out.println("Modulus = " + (a % b));
    }
}
```

Problems 4 | Output | Debug Console | Terminal | Ports

```
PS D:\Programs and Projects> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExc
eptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a65272379
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'Calculator'
Enter first number: 10
Enter second number: 25
Addition = 35
Subtraction = -15
Multiplication = 250
Division = 0
Modulus = 10
PS D:\Programs and Projects>
```

Task 4:
Write a program to check if a person is eligible to vote.
• Input: age and citizenship status (true or false).
• Use logical operators (&&, ||) to decide eligibility.

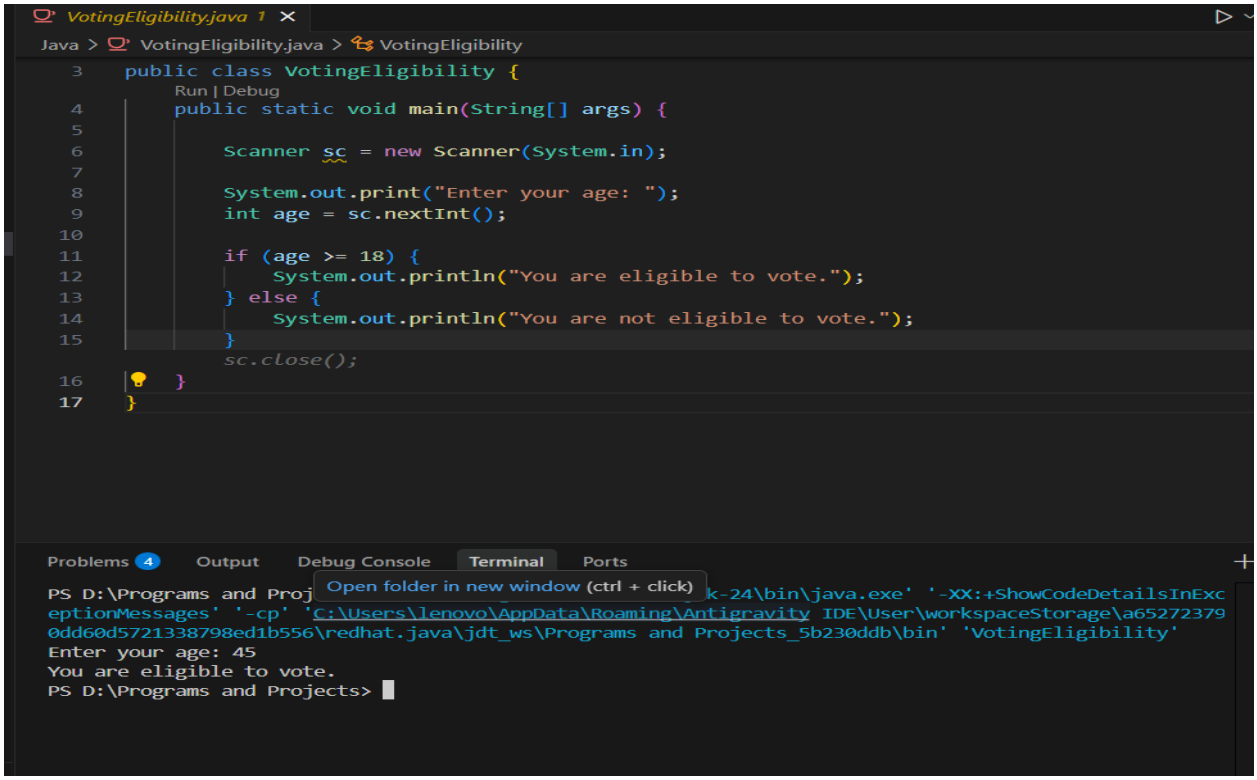
Solution :

```
public class VotingEligibility {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        if (age >= 18) {
            System.out.println("You are eligible to vote.");
        } else {
            System.out.println("You are not eligible to vote.");
        }
    }
}
```


VotingEligibility.java 1
Java > VotingEligibility.java > VotingEligibility
3 public class VotingEligibility {
Run | Debug
4 public static void main(String[] args) {
5
6 Scanner sc = new Scanner(System.in);
7
8 System.out.print("Enter your age: ");
9 int age = sc.nextInt();
10
11 if (age >= 18) {
12 System.out.println("You are eligible to vote.");
13 } else {
14 System.out.println("You are not eligible to vote.");
15 }
sc.close();
16 }
17 }
Problems 4
Output
Debug Console
Terminal
Ports
Open folder in new window (ctrl + click)
PS D:\Programs and Proj
k-24\bin\java.exe' '-XX:+ShowCodeDetailsInExc
eptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a65272379
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'VotingEligibility'
Enter your age: 45
You are eligible to vote.
PS D:\Programs and Projects>

Task 5:
Write a program that: • Takes two numbers and an operator symbol (+, -,*, /) from the user. Uses if-else statements to perform the appropriate operation.

Solution :

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result = " + (a + b));
        } else if (operator == '-') {
            System.out.println("Result = " + (a - b));
        } else if (operator == '*') {
            System.out.println("Result = " + (a * b));
        } else if (operator == '/') {
            System.out.println("Result = " + (a / b));
        } else {
            System.out.println("Invalid operator");
        }
    }
}
```

SimpleCalculator.java 1

Java > SimpleCalculator.java > ...

```java
import java.util.Scanner;

public class SimpleCalculator {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result = " + (a + b));
        } else if (operator == '-') {
            System.out.println("Result = " + (a - b));
        } else if (operator == '*') {
            System.out.println("Result = " + (a * b));
        } else if (operator == '/') {
            System.out.println("Result = " + (a / b));
        } else {
            System.out.println("Invalid operator");
        }
    }
}
```

Problems 4 | Output | Debug Console | Terminal | Ports

```
PS D:\Programs and Projects>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\le
novo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a652723790dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b2
30ddb\bin' 'SimpleCalculator'
Enter first number: 15
Enter second number: 7
Enter operator (+, -, *, /): /
Result = 2
PS D:\Programs and Projects>
```

Task 6:
Ask the user for a number and print its multiplication table from 1 to 10 using a for loop.

Solution :

```
import java.util.Scanner;

public class Table {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(n * i);
        }
    }
}
```

Table.java 1

Java > Table.java > Table

```java
import java.util.Scanner;

public class Table {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
    }
}
```

Problems 4 Output Debug Console Terminal Ports

```
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'Table'
Enter number: 15
15
30
45
60
75
90
105
120
135
150
PS D:\Programs and Projects>
```

Task 7:
Write a program that uses a while loop to find the sum of all even numbers between 1 and 50.

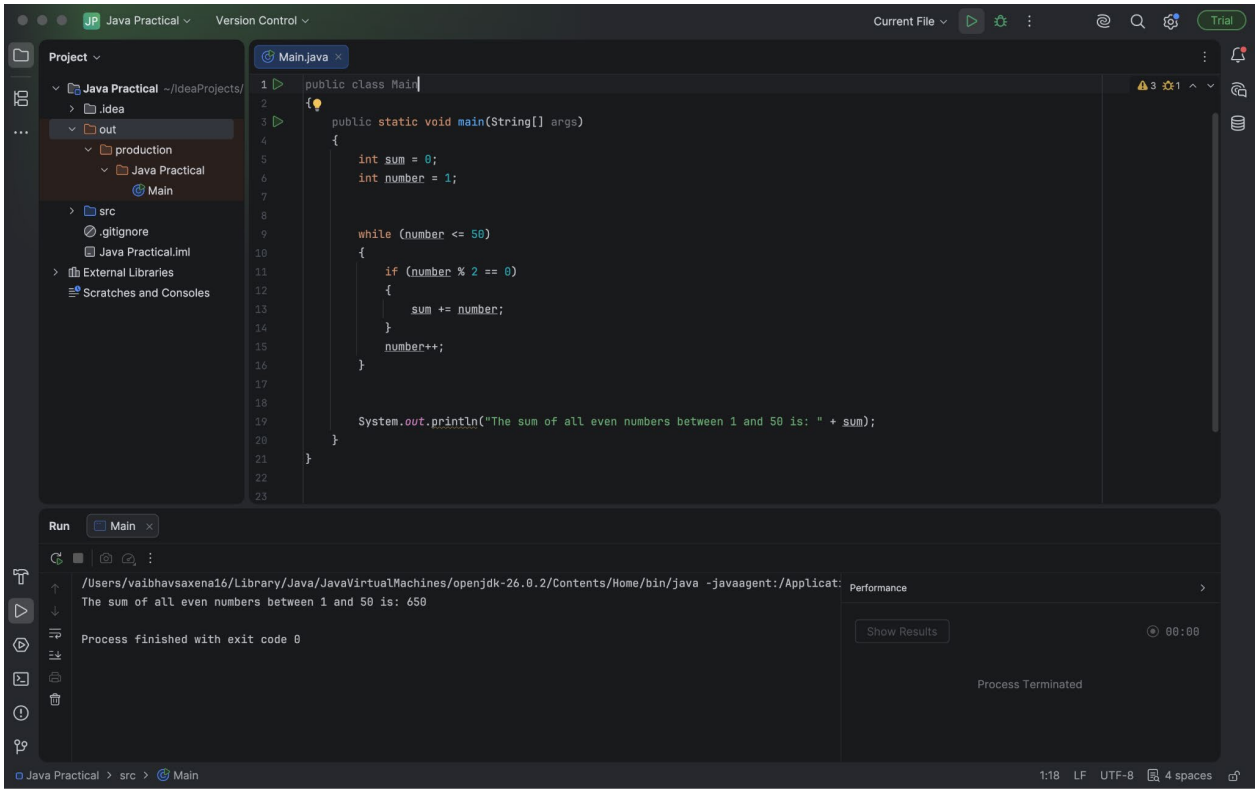
Solution :

```
public class SumOfEvenNumbers
{
   public static void main(String[] args)
{
        int sum = 0;
        int number = 1;

        while (number <= 50)
{
            if (number % 2 == 0)
{
                sum += number;
            }
            number++;
        }

        System.out.println("The sum of all even numbers between 1 and 50
is: " + sum);
    }
}
```

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Task 8:
Write a program that:
• Reads marks of 5 students into an integer array.
• Prints each student's marks.

Solution :

```
import java.util.Scanner;

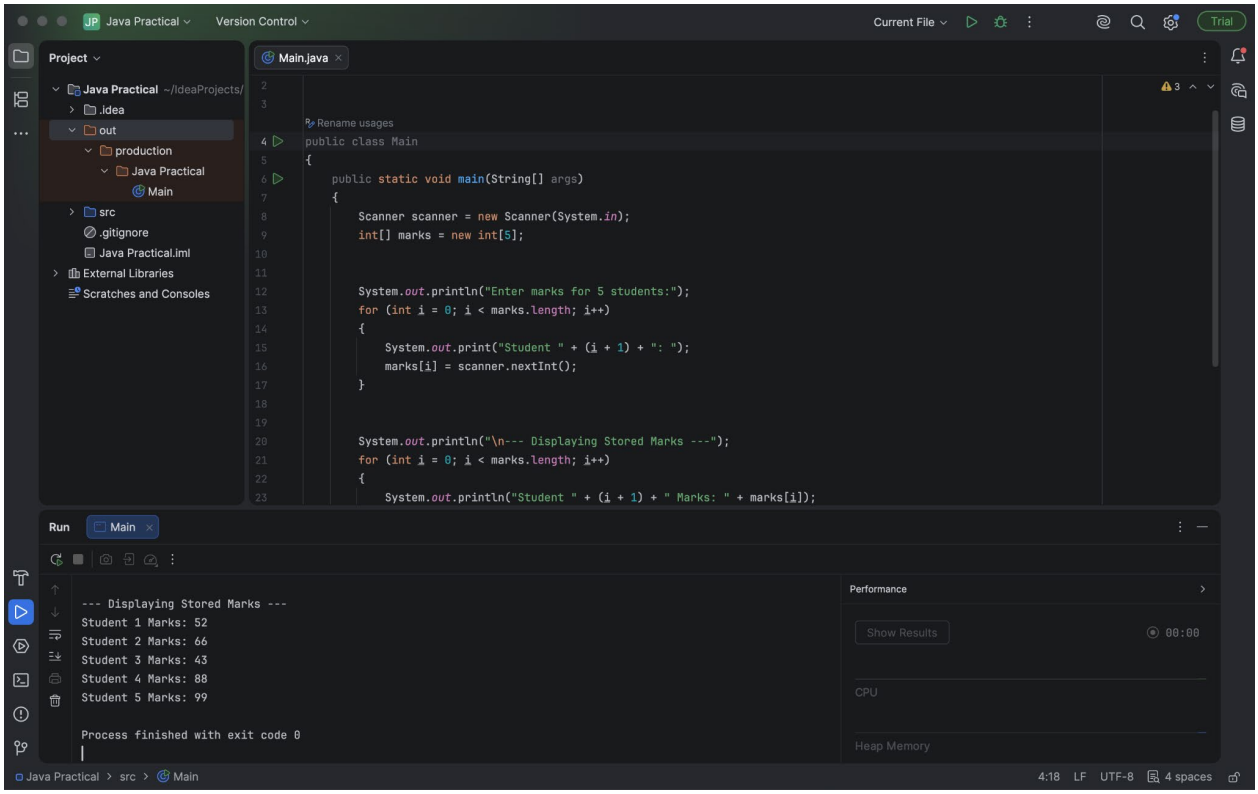
public class StudentMarksArray
{
   public static void main(String[] args)
   {
        Scanner scanner = new Scanner(System.in);
        int[] marks = new int[5];

        System.out.println("Enter marks for 5 students:");
        for (int i = 0; i < marks.length; i++)
        {
            System.out.print("Student " + (i + 1) + ": ");
            marks[i] = scanner.nextInt();
        }

        System.out.println("\n--- Displaying Stored Marks ---");
        for (int i = 0; i < marks.length; i++)
        {
            System.out.println("Student " + (i + 1) + " Marks: " +
marks[i]);
        }

        scanner.close();
   }
}
```

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Task 9:
Write a program to read 10 integers into an array and find the maximum value using a loop.

Solution :

```
import java.util.Scanner;

public class FindMaximumInArray
{
   public static void main(String[] args)
   {
        Scanner scanner = new Scanner(System.in);
        int[] numbers = new int[10];

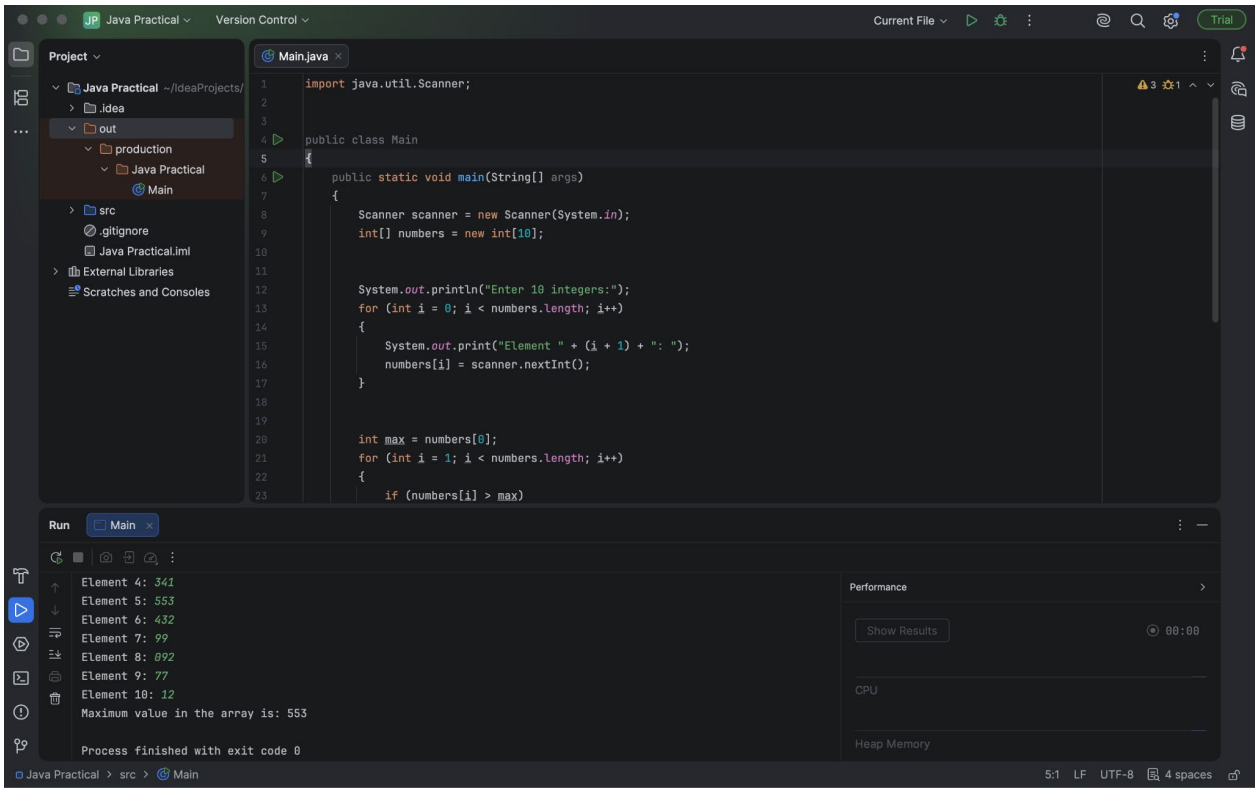
        System.out.println("Enter 10 integers:");
        for (int i = 0; i < numbers.length; i++)
        {
            System.out.print("Element " + (i + 1) + ": ");
            numbers[i] = scanner.nextInt();
        }

        int max = numbers[0];
        for (int i = 1; i < numbers.length; i++)
        {
            if (numbers[i] > max)
            {
                max = numbers[i];
            }
        }

        System.out.println("Maximum value in the array is: " + max);

        scanner.close();
   }
}
```

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Task 10:
Write a program that:
• Reads marks of 5 subjects into an array.
• Calculates the average marks.
• Uses if-else to assign grades:
o >= 90: A
o >= 75: Bo >= 50: C
o Else: Fail

Solution :

```
import java.util.Scanner;

public class StudentGradeSystem
{
   public static void main(String[] args)
   {
       Scanner scanner = new Scanner(System.in);
       double[] marks = new double[5];
       double totalSum = 0;

       System.out.println("Enter marks for 5 subjects:");
       for (int i = 0; i < marks.length; i++)
       {
           System.out.print("Subject " + (i + 1) + ": ");
           marks[i] = scanner.nextDouble();
           totalSum += marks[i];
       }

       double average = totalSum / 5.0;
       String grade;

       if (average >= 90)
       {
           grade = "A";
       } else if (average >= 75)
       {
           grade = "B";
```

```
        } else if (average >= 50)
        {
            grade = "C";
        } else {
            grade = "Fail";
        }

        System.out.println("\nAverage Marks: " + average);
        System.out.println("Grade: " + grade);

        scanner.close();
    }
}
```

